

ANNEXURE-IV

BCA (Honors) 4 Years Programme

Detailed Syllabus

Year - III

SEMESTER-V

**(To be implemented from Academic Year
2025-26 Onwards)**

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Major Discipline Specific Core BCAXXXX Fundamentals of Artificial Intelligence		Credits / Hours per week	4 (L-4, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Introduction to Artificial Intelligence Definitions and goals of artificial intelligence, Philosophy of AI, History of AI, limitations of symbolic knowledge representation through a production system or an expert system, Turing Test, Applications and impact of AI Problem Solving and Search Algorithms Introduction to AI problem-solving paradigms, Problem formulation and search space exploration, Uninformed search algorithms: depth-first search, breadth-first search, Informed search algorithms: A* search, Domain relaxations and heuristic functions Local Search and Evolutionary Algorithms: Hill climbing and simulated annealing, Genetic algorithms and genetic programming Adversarial Search: Minimax algorithm, Alpha-beta pruning and game tree search Advanced Search Algorithms: iterative deepening, bidirectional search		10	30%
2		Game Playing: Overview, And Example Domain: Overview, MiniMax, Alpha-Beta Cut-off, Refinements, Iterative deepening Knowledge Representation and Reasoning Propositional and predicate logic, Representation of knowledge using propositional logic, Importance of knowledge representation in AI, Semantic networks and frames, First-order logic and resolution, Prolog programming for logic-based reasoning Probabilistic Reasoning – Bayesian inference, Possible world representations, Heuristic reasoning methods. Blocks world Problem, Probabilistic reasoning and Bayesian networks, Fuzzy logic		15	30%

	and fuzzy systems		
3	Bayesian Networks Learning and Decision Theory Learning Bayesian networks from data Basics of planning in AI, State-space search and planning graphs. Decision theory: utility theory and decision networks Markov Decision Processes (MDPs): Introduction to MDPs, Value iteration and policy iteration algorithms Machine Learning Basics Introduction to machine learning, Supervised learning, unsupervised learning, and reinforcement learning, Evaluation metrics and cross-validation Introduction of reinforcement learning Ethical and Societal Implications of AI Bias and fairness in AI systems, Privacy concerns and data ethics, Responsible AI development and deployment	20	15%
4	Practical: Problems based on State Space Search & Heuristic Search Techniques, Game Playing, and Logic.	30	25%
Text Book			
1.	Artificial Intelligence: A Modern Approach, Stuart Russel, Peter Norvig (2022), PHI, 4 th Edition		
Reference Books			
1	Artificial Intelligence by Elaine Rich And Kevin Knight (2018), 2nd Edition, Tata Mcgraw-Hill		
2	Artificial Intelligence: A Guide for Thinking Humans, Melanie Mitchell (2019), Farrar, Straus and Giroux Publication		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Major Discipline Specific Core BCAXXXX: Design and Analysis of Algorithms		Credits / Hours per week	4 (L-4, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Analysis of Algorithm: Introduction, The efficient algorithm, Average, Best and worst case analysis, Amortized analysis, Asymptotic Notations, Analyzing control statement, Loop invariant and the correctness of the algorithm, Sorting Algorithms and analysis: Bubble sort, Selection sort, Insertion sort, Shell sort Heap sort, Sorting in linear time - Bucket sort, Radix sort and Counting sort Divide and Conquer Algorithm: Introduction, Recurrence and different methods to solve recurrence, Master Theorem, Multiplying large Integers Problem, Problem Solving using divide and conquer algorithm - Binary Search, Max-Min problem, Sorting (Merge Sort, Quick Sort), Matrix Multiplication.		15	33
2		Greedy Algorithm: General Characteristics of greedy algorithms, Problem solving using Activity selection problem, Elements of Greedy Strategy, Minimum Spanning trees (Kruskal’s algorithm, Prim’s algorithm), Graphs: Shortest paths, The Knapsack Problem, Job Scheduling Problem, Huffman code.		15	33
3		Dynamic Programming: Introduction, The Principle of Optimality, Problem Solving using Dynamic Programming: Calculating the Binomial Coefficient, Making Change Problem, Assembly Line-Scheduling, Knapsack problem, All Points Shortest path, Matrix chain multiplication, Longest Common Subsequence. Introduction to NP-Completeness: The class P and NP, Polynomial reduction, NP-Completeness Problem, NP-Hard Problems. Travelling Salesman problem, Hamiltonian problem		15	34
4		Practical: Implementation of various types of Algorithms and their applications in C and C++ • Divide and Conquer Algorithms		30	25

	• Greedy Algorithms • Dynamic Programming Algorithms • String Matching Algorithms • Matrix Multiplication Algorithms • Various Types of Tress and Graphs, their Traversal Techniques and Applications • Various Types of Searching, Sorting Algorithms and Their Applications.		
Text Book			
1	Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein (2009), PHI, 3 rd Edition		
Reference Books			
1	Fundamental of Algorithms, Gills Brassard, Paul Bratley (1996), PHI		
2	Introduction to Design and Analysis of Algorithms, AnanyLevitin (2017), Pearson, 3 rd Edition		
3	Foundations of Algorithms, Shailesh R Sathe (2014), Penram		
4	Design and Analysis of Algorithms, Dave and Dave (2013), Pearson, 2 nd Edition		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Major Discipline Specific Core BCAXXXX: Object-Oriented Software Engineering and Design Patterns	Credits / Hours per week	4 (L-3, P-1)	
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:	Maximum Marks / Grade	100	
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit	Contact Hours	Weightage(%)	
1		Introduction to Object-Oriented Analysis and Design Overview of Object-Oriented Paradigm, Importance of Object-Oriented Analysis and Design (OOAD) Unified Modeling Language (UML) Fundamentals Introduction to UML, Overview of Structural and Behavioral Diagrams in UML, Basic UML Diagrams: Class Diagrams, Use Case Diagrams, Sequence Diagrams, Understanding UML Notations and Symbols	15	33	
2		Use Case Diagram Understanding the purpose of Use Case Diagrams, Use Case Modeling, Actors and Use Cases, Relationships in Use Case Diagrams Class Diagram Understanding the basics of Class Diagrams, Classes, Attributes, and Methods, Associations, Aggregation, and Composition, Inheritance and Interfaces Sequence Diagram Introduction to Sequence Diagrams, Interaction between objects over time, Sequence Diagram notations and symbols Activity Diagrams Understanding the purpose and usage of Activity Diagrams, Modeling sequential processes using Activity Diagrams, Basic elements of Activity Diagrams: Activities, Actions, Control Flows, Decision Nodes, Merge Nodes, Using Swimlanes to represent actors or system components State Chart Diagram	15	33	

	Introduction to State Chart Diagrams, Modeling states and transitions, Hierarchy of states		
3	Introduction to Design Patterns Understanding Design Patterns, Creational Patterns: Singleton, Factory Method, Abstract Factory, Builder, Structural Patterns: Adapter, Composite, Decorator, Proxy, Behavioral Patterns: Observer, Strategy, Command, Iterator	15	34
4	Practical Students will be assigned group projects for Modeling software using Object Oriented Analysis and Design principles using UML diagrams, and Java with Design Patterns	30	25
Text Book			
1	Object-Oriented Software Engineering Using UML, Patterns, and Java by Bernd Bruegge, Allen Dutoit (2009), Pearson, 3 rd Edition		
Reference Books			
1	Craig Larman (2004) —Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Developmentll, Third Edition, Pearson Education		
2	Ali Bahrami(2017) - Object Oriented Systems Development - McGraw Hill International Edition		
3	Erich Gamma, and Richard Helm, Ralph Johnson, John Vlissides (1994) —Design patterns: Elements of Reusable Object-Oriented Softwareel, Addison-Wesley, 1 st Edition		
4	Martin Fowler(1999) —UML Distilled: A Brief Guide to the Standard Object Modeling Languageel, Third edition, Addison Wesley		
5	Fundamentals of Object Oriented Design in UML, Pearson Education by MeilirPage-Jones (2002), 1 st Edition		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- I BCAXXXX: Computer Graphics and Animation		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Basic of Computer Graphics: Basic of Computer Graphics, Applications of computer graphics, Display devices, Random and Raster scan systems, Graphics input devices, Graphics software and Standards Graphics Primitives: Points, lines, circles and ellipses as primitives, scan conversion algorithms for primitives, Fill area primitives including scan-line polygon filling, inside-outside test, boundary and flood-fill, character generation, line attributes, area-fill attributes, character attributers.		15	25%
2		2D transformation and viewing: Transformations (translation, rotation, scaling), matrix representation, homogeneous coordinates, composite transformations, reflection and shearing, viewing pipeline and coordinates system, window-to-view port transformation, clipping including point clipping, line clipping (cohen-sutherland, liang- bersky, NLN), polygon clipping		15	25%
3		3D concepts and object representation: 3D display methods, polygon surfaces, tables, equations, meshes, curved lies and surfaces, quadric surfaces, spline representation, cubic spline interpolation methods, Bazier curves and surfaces, B-spline curves and surfaces		15	25%
4		Practicals: - To draw a line using DDA Algorithm for positive line slope. Also for negative line. - To draw a line using Bresenham’s Algorithm for positive line slope. -To display simple shapes (Like hut, star, car etc.) Using graphics primitives -To perform translation transformation on a given triangle. -To perform scaling transformation on a given triangle. -To perform rotation transformation on a given triangle. -To rotate a given triangle clockwise and anticlockwise about a given point.		30	25%

	-To perform reflection of a point about a line $y= mx+c$.		
Text Book			
	Computer Graphics, D.Hearn, P.Baker-C Version, Pearson, 2 nd edition, 2002.		
Reference Books			
1	Computer Graphics, with OpenGLHearn and Baker, Pearson, 4 th edition, 2011.		
2	Computer Graphics, Sinha & Udai,-TMH		
3	Computer Graphics, Foley and vanDam-PersonEducation,1 st edition,2013.		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- I BCAXXXX: Software Project Management		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Introduction – Overview of a Project, Project types and operations, Program, Project management concepts – Characteristics and Goals of a Project – Project Execution and Stakeholders – Project Management with its cross functional nature – 4P’s for effective Project Management – Principles of Project management – Software Project Life cycle phases Project Management and Control: Project planning, scheduling, Organizational Structures, The Project Organization, The Functional Organization, The Matrix organization, designing an Organization, Building the Team, Leadership. Project tracking – Change control – Software Configuration Management.		15	25%
2		Software Process, Project Metrics and Quality: Overview of Measures, Metrics, And Indicators – Metrics in the Process And Project Domains – Software Measurement – Metrics For Software Quality – Project Planning Objectives - Software Scope: Obtaining necessary information and Feasibility – Resources - Software Quality Assurance - Software Reviews - Format Technical Review (FTR)		15	25%
3		Software Project Estimation and Risk Management - What is Software Project Estimation? – Decomposition techniques: Software sizing, Problem based estimation, Process based Estimation – Empirical estimation models: COCOMO model – Concept of Risk, Type of Risks – Risk Identification with Components and drivers – Risk Assessment – Risk Mitigation, Monitoring and Management		15	25%
4		Practicals: - Create project plan for given system - Generate project report and tool - Make SPMP for given system - Cocomo model related exercise - FP based exercise - Critical path calculation		30	25%

	- Cost reports (Earned value reports,Task cost report) - Project management tool introduction and use(MS project or any)		
Text Book			
1	McGraw Hill Higher Education; 5th ed. edition (1 December 2001)		
Reference Books			
1	Walker Royce, “Software Project Management – A Unified Framework”, 2001, Pearson Education		
2	S.A.Kelkar, “Software Project Management – A concise study”, 2nd Ed. PHI learning Pvt. Ltd. 2009		
3	Ian Sommerville, “Software Engineering”, 2004, Addison Wesley.		
4	PankajJalote “An Integrated Approach to Software Engineering” 2nd Edition.		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- I BCAXXXX: Microprocessor and Interfacing		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Introduction to Microprocessor, Components of a Microprocessor: Registers, ALU and control & timing, System bus (data, address and control bus), Microprocessor systems with bus organization. Microprocessor Architecture and Operations, Memory, I/O devices, Memory and I/O operations. 8085 Microprocessor Architecture, Address, Data and Control Buses, 8085. Pin Functions, Demultiplexing of Buses, Generation of Control Signals, Instruction Cycle, Machine Cycles, T-States, Memory Interfacing.		18	30%
2		Assembly Language Programming Basics, Classification of Instructions, Addressing Modes, 8085 Instruction Set, Instruction and Data Formats, Writing, Assembling & Executing a Program, Debugging The Programs. Writing 8085 assembly language programs with decision, making and looping using data transfer, arithmetic, logical and branch instructions. Stack & Subroutines, BCD Arithmetic and 16-Bit Data operations.		18	30%
3		Interfacing Concepts, Ports, Interfacing Of I/O Devices, Interrupts In 8085.		9	15%
4		Practical: 1. Write an 8085 assembly language program for exchanging two 8-bit numbers stored in memory locations 2050h and 2051h. 2. Write an 8085 assembly language program to add two 8-bit numbers stored in memory locations 2050h and 2051h. Store result in location 2052h. 3. Write an 8085 assembly language program to add two 16-bit numbers stored in memory. 4. Write an 8085 assembly language program to add two decimal numbers using DAA instruction.		30	25%

	5. Write an 8085 assembly language program to find the minimum from two 8-bit numbers. 6. Write an 8085 assembly language program to get the minimum from block of N 8-bit numbers. 7. Write an 8085 assembly language program to add block of 8-bit numbers. 8. Write an 8085 assembly language program to find the number of 1's binary representation of given 8-bit number. 9. Write an 8085 assembly language program to count the length of string ended with 00h starting from location 2050h. 10. Write an 8085 assembly language program to compute even parity and insert it as MSB in 8-bit number. 11. Write a subroutine to exchange two8-bit numbers. Use it to reverse an array of 8-bit numbers.		
Text Book			
1	Microprocessor Architecture, Programming, and Applications with the 8085, Ramesh S. Gaonkar (2002) Pub: Penram International., 5 th Edition		
Reference Books			
1	Microprocessors and Interfacing, N. Senthil Kumar, M. Saravanan, S. Jeevanathan, S. K. Shah (2012), Oxford		

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Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- I BCAXXXX: Cryptography and Network Security		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Security Concepts: Introduction, The need for security, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security Cryptography Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks.		15	25%
2		Symmetric key Ciphers: Block Cipher principles, DES, AES, Blowfish, RC5, IDEA, Block cipher operation, Stream ciphers, RC4. Asymmetric key Ciphers: Principles of public key cryptosystems, RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, Knapsack Algorithm. Cryptographic Hash Functions: Message Authentication, Secure Hash Algorithm (SHA-512) Message authentication codes: Authentication requirements, HMAC, CMAC, Digital signatures, Elgamal Digital Signature Scheme.		15	25%
3		Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption, Distribution of Public Keys, Kerberos, X.509 Authentication Service, Public – Key Infrastructure Transport-level Security: Web security considerations, Secure Socket Layer and Transport Layer Security, HTTPS, Secure Shell (SSH) Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN, IEEE 802.11i Wireless LAN Security E-Mail Security: Pretty Good Privacy, S/MIME IP Security: IP Security overview, IP Security architecture, Authentication Header, Encapsulating security payload, Combining security associations, Internet Key Exchange		15	25%
4		Practical: - Implement Caesar cipher encryption-decryption.		30s	25%

	- Implement Monoalphabetic cipher encryption-decryption. - Implement Playfair cipher encryption-decryption. - Implement Polyalphabetic cipher encryption-decryption. - Implement Hill cipher encryption-decryption. - To implement Simple DES or AES. - Implement Diffi-Hellmen Key exchange Method. - Implement RSA encryption-decryption algorithm. - Write a program to generate SHA-1 hash. - Implement a digital signature algorithm. - Perform various encryption-decryption techniques with cryptool. - Study and use the Wireshark for the various network protocols. Design based problems: - Study the standard document for the security policy for an organization and prepare the detailed security policy document for managing information security for your institute. - Study the keytool provided by the Java to generate key pairs for public key cryptography. Design and develop your own such tool to generate the key pair and test the pair with RSA implementation for encryption-decryption. - Study how the browsers manage the digital certificates for various secured websites for making secured communication.		
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Text Book

1	Cryptography and Network Security - Principles and Practice: William Stallings (2022), Pearson Education, 6th Edition
2	Cryptography and Network Security: Atul Kahate (2019), McGraw Hill, 4th Edition

Reference Books

1	Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan (2011), Wiley India, 1st Edition.
2	Cryptography and Network Security: Forouzan Mukhopadhyay, McGraw Hill, 3rd Edition
3	Information Security, Principles, and Practice: Mark Stamp (2021), Wiley India, 3 rd Edition
4	Principles of Computer Security: WM. Arthur Conklin (2015), Greg White, TMH, 4 th Edition
5	Introduction to Network Security: Neal Krawetz (2007), CENGAGE Learning
6	Network Security and Cryptography: Bernard Menezes (2018), CENGAGE Learning, 1 st Edition

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- I BCAXXXX: Linux System Administration		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Introduction to Linux - Overview of Linux operating system, History and development of Linux, Understanding Linux distributions, Introduction to command line interface (CLI), Basic Linux commands and file system navigation Linux File System and Permissions - Understanding Linux file system hierarchy, File and directory permissions in Linux, Managing users and groups, Access control lists (ACLs) in Linux, Practical exercises on file system management and permissions Package Management - Introduction to package management systems (APT, YUM, DNF), Installing, updating, and removing software packages, Managing software repositories, Practical exercises on package management Process Management - Understanding processes and process lifecycle in Linux, Process management commands (ps, top, kill, etc.), Managing process priorities and scheduling, Practical exercises on process management		15	35%
2		System Monitoring and Performance Tuning - Monitoring system performance using built-in tools (vmstat, iostat, sar, etc.), Analyzing system logs, Performance tuning techniques, Practical exercises on system monitoring and performance tuning Networking - Understanding networking concepts in Linux, Configuring network interfaces, Managing network services (DHCP, DNS, SSH, etc.), Firewall management using iptables or firewalld, Practical exercises on networking configuration Storage Management - Disk partitioning and formatting, Managing file systems (ext4, XFS, etc.), Logical Volume Manager (LVM) for flexible storage management, RAID configuration and management, Practical exercises on storage		15	30%

	management		
	Backup and Recovery - Implementing backup strategies (full, incremental, differential), Using backup tools (tar, rsync, dd, etc.), Disaster recovery planning, Practical exercises on backup and recovery		
3	Security - Understanding Linux security principles, User authentication methods (passwords, SSH keys, etc.), File system security (permissions, SELinux, AppArmor), Intrusion detection and prevention, Practical exercises on Linux security Automation and Scripting - Introduction to shell scripting (Bash) Writing and executing shell scripts, Automating system administration tasks, Using cron for scheduling tasks, Practical exercises on shell scripting and automation Virtualization and Containers Introduction to virtualization (KVM, VirtualBox, etc.), Managing virtual machines, Introduction to containerization (Docker, Podman), Practical exercises on virtualization and containers	15	35%
4	Practical: • Basic Linux Utilities • Linux File System and Permissions • Package Management • Process Management • System Monitoring and Performance Tuning • Networking • Storage Management • Backup and Recovery • Linux Security • Linux Shell Scripting and Automation • Virtualization and Containers	30	25%
Text Book			
1	Practical Linux System Administration by Kenneth Hess (2023), O'Reilly Media, Inc.		
Reference Books			
1	UNIX and Linux System Administration Handbook, 5th Edition by Trent R. Hein, Evi Nemeth, Garth Snyder, Ben Whaley, Dan Mackin, Addison-Wesley Professional		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- I BCAXXXX: Software Architecture		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Introduction: Overview of Software development methodology and software quality model, different models of software development and their issues. Introduction to software architecture, evolution of software architecture, software components and connectors, common software architecture frameworks, Architecture business cycle – architectural patterns – reference model. Software architecture models: structural models, framework models, dynamic models, process models. Architectures styles: dataflow architecture, pipes and filters architecture, call-and return architecture, data-centered architecture, layered architecture, agent based architecture, Micro-services architecture, Reactive Architecture, Representational state transfer architecture etc.		15	25%
2		Design Pattern Catalog and Behavioral Patterns - Structural patterns, Adapter, bridge, composite, decorator, facade, flyweight, proxy. Behavioral Patterns: Chain of Responsibility, Command, Interpreter, Iterator, Mediator, Memento, Observer, State, Template Method, Singleton Design pattern		15	25%
3		Interactive systems and the MVC architecture - Introduction, The MVC architectural pattern, analyzing a simple drawing program, designing the system, designing of the subsystems, getting into implementation, implementing undo operation, drawing incomplete items, adding a new feature, pattern-based solutions. Service Oriented Architecture, Microservices and Cloud Native Architectures		15	25%
4		Practical on • Explain Structural Models. • Explain Framework Models. • Explain Dynamic Models. • Explain Process Models.		30	25%

	• Explain Attribute Driven Design method (ADD). • Explain different patterns.		
Text Book			
1	Dikel, David, D. Kane, and J. Wilson, “Software Architecture: Organizational Principles and Practices”, Prentic -Hall.		
Reference Books			
1	Frank Bachmann, RegineMeunier, Hans Rohnert “Pattern Oriented Software Architecture” –Volume 1, 1996.		
2	William J Brown et al., "Anti-Patterns: Refactoring Software, Architectures and Projects in Crisis", John Wiley, 1998.		
3	Albin, S. “The Art of Software Architecture”, Indiana: Wiley, 2003.		
4	Erich Gamma, Richard Helan, Ralph Johman, John Vlissides, Design Patterns, Pearson Publication,2013.		
5	Bass, L., P. Clements, & R. Kazman, “Software Architecture in Practice”, 2 Ed, Prentice-Hall.		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- I BCAXXXX: Augmented Reality and Virtual Reality		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Introduction, Components of a VR system, 3D User Interface Input and Output devices, 3D viewing, Designing & Building VR Systems, Introduction to Augmented Reality (AR), VR Modeling: Geometric modeling, Kinematic, Physical and Behavior modeling; Selection and Manipulation during 3D Interaction		7	10%
2		Virtual Environment Generic VR system: Introduction, Virtual environment, Computer environment, VR technology, Model of interaction, VR Systems, Animating the Virtual Environment: Introduction, The dynamics of numbers, Linear and Nonlinear interpolation, the animation of objects, linear and non-linear translation, shape & object in between, free from deformation, particle system Physical Simulation: Introduction, Objects falling in a gravitational field, Rotating wheels, Elastic collisions, projectiles, simple pendulum, springs, Flight dynamics of an aircraft.		18	40%
3		Augmented Reality (AR) Taxonomy, Technology and Features of Augmented Reality, AR Vs VR, Challenges with AR, AR systems and functionality, Augmented Reality Methods, Visualization, Techniques for Augmented Reality, Enhancing interactivity in AR Environments, Evaluating ARsystems, Travel and Wayfinding in Virtual Environments, Strategies for Designing and Developing 3D UIs, Evaluation of 3D User Interfaces, Traditional and Emerging VR/AR applications Human Factors in Virtual Reality, Case study on Construction of Geographic Virtual World. Group assignments on implementation of a Virtual/ Augmented Reality Application using open-source toolkits/ libraries such as OpenSceneGraph, Vega, VRML etc.		20	50%
4		Practical: 1. Study of different game engines		30	25%

	2. Implementation on Video/ Feature Viewing 3. Implementation on Virtual tour 4. Implementation on material animation 5. Implementation to show portal planets 6. Explore projects in Unity 2D and 3D 7. Mini Project on Augmented Reality or Virtual Reality		
Text Book			
1	Coiffet, P., Burdea, G. C., (2003), “Virtual Reality Technology,” Wiley-IEEE Press, ISBN: 9780471360896		
2	Schmalstieg, D., Höllerer, T., (2016), “Augmented Reality: Principles & Practice,” Pearson, ISBN: 9789332578494		
3	G J Kim, “Designing VR Systems: The Structured Approach”, Springer.		
Reference Books			
1	D.A. Bowman et al.(2017), “3D User Interfaces: Theory and Practice”, Addison Wesley, 2 nd Edition		
2	John Vince (2002), “Virtual Reality Systems”, Pearson Ed., 1 st Edition		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- I BCAXXXX: Agile Methodologies		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Introduction to Agile Methodologies Overview of Agile principles and values, History and evolution of Agile, Agile vs. traditional project management, Agile Manifesto and its significance Scrum Framework Scrum roles (Product Owner, Scrum Master, Development Team), Scrum artifacts (Product Backlog, Sprint Backlog), Scrum ceremonies (Sprint Planning, Daily Standup, Sprint Review, Sprint Retrospective), Scrum implementation and best practices Kanban Methodology Introduction to Kanban principles, Kanban boards and visualization, Work-in-progress (WIP) limits, Continuous improvement with Kanban		15	33
2		User Stories and Product Backlog Writing effective user stories Prioritizing and managing the product backlog Story mapping and backlog grooming Estimation techniques (e.g., story points) Agile Planning and Sprints Sprint planning and commitment, Sprint execution and daily standup, Sprint review and demonstration, Sprint retrospective and continuous improvement Agile Metrics and Reporting Velocity and burndown charts, Cycle time and lead time, Agile reporting and dashboards, Using metrics for decision-making		15	33
3		Scaling Agile Agile at scale (e.g., SAFe, LeSS), Distributed Agile teams and challenges, Agile transformation strategies, Case studies of large-scale Agile adoption Agile Challenges and Best Practices		15	34

	Common Agile challenges and solutions, Agile in regulated industries, Balancing flexibility and predictability, Achieving organizational agility Agile Future Trends Emerging trends and the future of Agile Group Agile Project Collaborative group project applying Agile principles, Agile project planning and execution, Demonstrating Agile practices in project work		
4	Practicals: - Practical Lab with Atlassian JIRA Agile practices in project work	30	25%
Text Book			
1	Agile Estimating and Planning, Mike Cohn, Pearson, 2005		
Reference Books			
1	Scrum: The Art of Doing Twice the Work in Half the Time, Jeff Sutherland, Currency, 2014		
2	Kanban: Successful Evolutionary Change for Your Technology Business, David J. Anderson, Blue Hole Press, Blue Book ed. Edition, 2010		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- I BCAXXXX: Software Testing And Quality Assurance		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Software Testing: Introduction, Types of Software ▪ Benefits of Software Testing ▪ Manual Testing ▪ Automation Testing ▪ Types of Testing ▪ Functional Testing. Introduction to software Testing Levels: Software Testing Levels ▪ Unit Testing ▪ Integration Testing ▪ System Testing ▪ Acceptance Testing ▪ Performance Testing ▪ Security Testing ▪ Maintenance Testing ▪ Regression Testing Compatibility Testing: Overview, Backward and Forward Compatibility. Testing Multiple versions. Data Sharing Compatibility Testing Concepts : Test Scenarios ▪ RTM ▪ Test Log ▪ Test Summary, Test Cases, Test Plan and Key Components ▪ Test Plan ▪ Test Plan and Concepts Key Components ▪ Attributes of Test Plan as per IEEE 829 Format		10	10%
2		Defect Life Cycle , Defect Report and Jira : What is Quality? ▪ Difference between QA and QC ▪ What is PDCA cycle or Deming cycle ▪ What is a Defect? ▪ Difference between Failure, Error, Bug and Defect ▪ What is a Failure? ▪ What is a Error? ▪ Real-Time Example of a Failure ▪ Defect Life Cycle ▪ Information on Defect Report. Software Test Automation: What is Test Automation, Terms used in automation, Skills needed for automation, what to automate, scope of automation, Challenges in Automation, Data-driven Testing. Automation Tools like JUnit, Jmeter		15	15%
3		Selenium Tool: Introducing Selenium, Brief History of The Selenium Project, Selenium’s Tool Suite, SeleniumIDE, Selenium RC, Selenium Web driver, Selenium Grid, Test Design Considerations Quality Assurance: Testing, QA and QC, Quality Management, Quality Planning Process, Quality Assurance Process, Quality Control process Organisational Structures: CMM Capability Maturity		20	25%
4		Practical: Creating tests scenarios, test cases, defect report, and test plan for both mobile app		15	25%

	as well as web app. Web Site Testing: Web Page Fundamentals, Black Box Testing: Text, Hyperlinks, graphics, Forms. Gray Box Testing & White Box Testing, Configuration and Compatibility Testing - Jira Tool: Install and Configure Jira Tool ▪ How to add Users in Jira? ▪ How to create backlogs? ▪ How to create a story What is Zephyr in Jira Platform ▪ How to create Test Case?		
1	Kshirasagar Naik and Priyadarshi Tripathy, Software Testing and Quality Assurance: Theory and Practice, John Wiley & Sons, Inc., 2008		
2	William Perry, “Effective Methods for Software Testing”, John Wiley & Sons, New York, 1995.		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- II BCAXXXX: Advanced Python Programming		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage (%)
1		Object oriented Programming with Python Basics of class, object and instance. Class level attributes and instance level attribute. Constructor and other magic methods. Bound and unbound methods. Built in functions for python class and objects. Database Programming using Python Introduction to MySQL, PYMYSQL Connection, Executing queries, Transactions • Creating DB Table • INSERT, READ, UPDATE, DELETE operations • COMMIT & ROLLBACK operation handling Errors		15	33
2		GUI Programming using Python Introduction, Python’s GUI library: Tkinter (installing and getting started), Introduction to GUI programming, top-level window, Tk widgets: Label, Button, Entry, Frame, Checkbutton, Radiobutton, Combobox, Scale, ProgressBar, Developing GUI based applications using Tkinter. Python libraries and Data collection Introduction to Numpy, Numpy Array, Numpy Mathematical Function, Numpy Trigonometric Function, Numpy random module,Numpy String Operation.Pandas, Pandas data structure(series and dataframe),Data Wrangling using pandas>Loading dataset in to a data frame, Selecting columns from a data frame, Adding new data in a data frame, Deleting data from a data frame)Introduction to Matplotlib, Scatterplot, Line Plot, Bar Chart,Histogram, Box plot, Violin plot		15	33
3		Python for Model Development and Evaluation ML: Introduction to machine learning- Supervised and Unsupervised Learning, Model development using Linear Regression, Model Visualization, Prediction and Decision Making, Model Evaluation: Over-fitting, Under-fitting and Model		15	34

	Selection DIP: Introduction to digital image Processing, Basic function to import draw and display images using Python.		
4	Practical: Practicals based on Matplotlib, Pandas, Numpy, Developing GUI based applications using Tkinter 1. Write a program to Python program to implement concepts of OOP such as a. Types of Methods b. Inheritance c. Polymorphism 2. Write a program to Python program to implement concepts of OOP such as a. Abstract methods and classes b. Interfaces 3. Write a Python program demonstrating the use of box and violin plot. 4. Write Python programming to create a pie chart with a title of the popularity of programming Languages. 5. Write a Python program to draw a scatter plot for three different groups comparing weights and heights. 6. Write a Python program to create a table and insert some records in that table. Finally select all rows from the table and display the records.	30	25
Text Book			
1	Reema Thareja,Python Programming: Using Problem Solving Approach, Oxford University Press, 2023		
2	Building Modern GUIs with tkinter and Python: Saurabh Chandrakar, Dr. Nilesh Bhaskarrao Bahadure BPB Publications, 27 June 2023		
3	Wes McKinney,Python for Data Analysis: Data Wrangling with Pandas, NumPy, and jupyter 3 rd Edition , O'Reilly Media, 20 September 2022		
Reference Books:			
1	Learn Programming in Python with Cody Jackson, Cody Jackson, Packt Publishing, 30 November 2018		
2	Python Basics: A Practical Introduction to Python 3, David Amos, Dan Bader , Joanna Jablonski, Fletcher Heisler , March 16, 2021		
3	Introduction to Computing and Problem Solving using Python, E. Balagurusamy, McGraw Hill Education India pvt ltd, New Delhi, 1 July 2017		
4	Beginning Python : Using Python 2.6 and Python 3.1, James Payne, Wiley, January 2010		
5	Think Python : How to Think Like a Computer Scientist, Allen Downey, O'Reilly, USA, 28 December 2015		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective-II BCAXXXX: Advanced Java Programming		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Java Networking & JDBC API Network Basics and Socket overview, TCP/IP client sockets, URL, TCP/IP server sockets, Datagrams, java.net package Socket, ServerSocket, InetAddress, URL, URLConnection, The JDBC Connectivity Model, Database Programming: Connecting to the Database,Creating a SQL Query, Getting the Results, Updating Database Data, Error Checking and the SQLException Class, The SQLWarning Class, The Statement Interface,PreparedStatement, CallableStatement The ResultSet Interface, Updatable Result Sets, JDBC Types, Executing SQL Queries, ResultSetMetaData, Executing SQL Updates, SQL 99 datatypes,ConnectionPooling,RowSet,TransactionManagement,JDBC 4.0 features,Working with properties file		20	25%
2		Servlet API Servlet Model: Overview of Servlet, Servlet Life Cycle, HTTP Methods Structure and Deployment descriptor ServletContext and ServletConfig interface, Attributes in Servlet, Request Dispatcher interface, The Filter API: Filter, FilterChain, Filter Config Cookies and Session Management: Understanding state and session, Understanding Session Timeout and Session Tracking, URL Rewriting,Listeners Authentication and Authorization(Security In Servlet Programming),Working with Servlet 2.5/3.0/3.1 features,Annotations in Servlet Programming Java Server Pages JSP Overview: The Problem with Servlets, Life Cycle of JSP Page, JSP Processing, JSP Application Design with MVC, Setting Up the JSP Environment, JSP Directives, JSP Action,		25	25%

	JSP Implicit Objects JSP Form Processing, JSP Session and Cookies Handling, JSP Session Tracking JSP Database Access, JSP Standard Tag Libraries, JSP Custom Tag, JSP Expression Language, JSP Exception Handling, JSP XML Processing,JSTL,MVC Architecture		
3	Java Frameworks:-Hibernate 5.0 Issues with Persistence layers and Object-Relational Mapping (ORM) ,Hibernate Overview and Benefitsof Hibernate, POJO (Plain Old Java Object) Based Mapping, Configuring Hibernate (hibernate.cfg.xml) file Hibernate Mapping,File, Mapping the Entity Class,Primary key generation strategies,HQL – Hibernate Query Language, Transaction Overview and Transactions in Hibernate, The lifecycle of managed objects, Inheritance Mapping	25	25%
4	Lab Assignments : Programs using JDBC API Programs using Servlet API Programs using JSP - Programs using Hibernnet framework	30	25%
Text Book			
1	Complete Reference J2EE by James Keogh Mcgraw publication, 1 st Edition, 2017		
Reference Books			
1	Professional Java Server Programming by SubrahmanyamAllamaraju, Cedric Buest Wiley Publication, 1.3 Edition, 2001		
2	Java Persistence with Hibernate by Christian Bauer, Gavin King, 2 nd Edition, 2015		
3	JDBC™ API Tutorial and Reference, Third Edition, Maydene Fisher, Jon Ellis, Jonathan Bruce, Addison Wesley, Subsequent Edition, 2003		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective-II BCAXXXX: Advanced Programming with.NET Framework and C#		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Web Application Development The base Class Library, CLR, CTS, Common Language Specification, Namespaces, Assemblies, Application Domains. Developing Web applications using ASP.net - The code-behind programming model, ASP.net Web Forms creating user controls, server controls, and server-side data access; ASP.net data access, State Management in ASP.NET, .aspx and .aspx.cs files, a configuration file Web.config; User Authentication and Authorization. Developing user controls and custom controls		20	25%
2		The .NET Distributed Architecture Remote Processing, World Wide Web and the .Net Solution. Messaging - Messaging Architecture, setting up Messaging Network, Creating and Referencing Message Queues, Sending and receiving Messages, Message Queue Security, Remoting - Remoting Architecture, Instantiating Using Activator Methods: GetObject() and CreateInstance(), Communicating From the Remote Object Using Events, Asynchronous Calls; Transactions - Manual Transactions and automatic Transactions. Networking in .NET: System.NET Namespace, The Web Client Class, HTTP Derivations of Web Request and Web Response, TCP and Socket Classes		25	25%

3	Class Parsing XML with the Xml Namespace and Web Services Parsing XML - XmlReader Classes, XmlDocument Class; XMLDOM and the XmlDocument class, Modifying an XmlDocument.XML Web services overview, understanding the SOAP and WSDL, creating and using a Web service, state management for XML Web services. .NET Security Socket Code Access Security, Web Service Security. Assemblies and deployment The manifest, private assemblies and shared assemblies, Global Assembly Cache, Deploying .Net windows and web applications, VS.NET Build Configurations. Introduction to Windows Presentation Foundation (WPF), Windows Communication Foundation (WCF), Windows Workflow Foundation (WF), Introduction to MVC framework, .Net Core.	25	25%
4	Practicals: - Create a form in asp.net -Validation control -Working with ado.net in asp.net -Working with Master pages and themes -Ad-rotator control, custom and user control - Creating web services in asp.net - State management in web service - Using XmlReader, XmlDocument class, Modifying Xmlldocument -Creating assemblies (exe and dll) - Working with WPF - -Working with WCF	30	25%

Text Book

1	Professional C# .Net - by Christian Nagel, (2021 edition) - Wrox Publication
2	Advanced .Net Technology- by Chirag A.Patel (2011) - Dreamtech Press

Reference Books

1	Professional C# 4 and .NET 4 – by Christian Nagel, BillEvjen, Jay Glynn, karli Watson, Morgan Skinner(2010): Wrox Publication
2	Beginning ASP.NET 4.5 in C# - by Macdonald Matthew(2012): Apress
3	Professional C# 6 and .NET Core 1.0 – by Christian Nagel (2016) : Wrox publication
4	Pro ASP.NET MVC 5 – by Freeman, Adam : (5 th edition ,2014) : Apress

		The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications		Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- II BCAXXXX: Programming with Go		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Introduction TO GO History, features, Advantages, Disadvantages, Installation in different OS,Datatypes,Package,Control statement(like : if-else,switch,for,for range, Goto,Break)		10	30%
2		Function, Array and String: Go Function, Recursion, Closure, Array:Go array, Slice, command args, Go String		15	30%
3		Go Types, GO Map and Error Go Interface, Go Pointer,Go Reflect, Go Rune, Go Map,Go Error, Go Recover, Go Defer, Go Panic Go Concurrency, Go Time and Miscellaneous Topics Go Concurrency, Go Race, Go Mutex, Go Atomic Variable, Go Channel, Go Worker Pools, Go Time, Go Epoch, Go Tickers, Go File I/O,Go HTTP Server, Go URL Parsing, Go Rest API, Go Base64 Encoding, Go Random Number, Go Sorting, Go JSON		20	40%
4		Practical Functions Array, strings, args Map, error, recover - Concurrency, Mutex, Parsing, JSON		30	25%
Text Book					
1		Introducing Go: Build Reliable, Scalable Programs by Caleb Doxsey, 1 st Edition, 2016			
2		The Go Programming Language by Alan A.A. Donovan and Brian W. Kernighan, 2016			
Reference Books					
1	Get Programming with Go by Nathan Youngman, Roger Peppé, Manning				
Web Resources					
1	https://www.golang-book.com/books/intro				

		The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications		Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- II BCAXXXX: Programming with Rust		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Rust Basics: Variables and Mutability, Scalar Data types, Inferred types, Unused variables, Interior Mutability, Mutable references, Constants, constfn, Associated constants, Modules and Packages, Nesting modules, Drop, Copy, Operator Overloading, Cargo, Importing-Exporting items, crate::,super::,self::,External crates, Renaming Imports, Macros(Declarative and Procedural), Function-like Macros, Panicking Macros, Inner-Outer Attributes, Type Signature, Shadowing, Pattern binding, Statements and Expressions, Tuples, Arrays, Vectors, Slices, String, Structs, Methods, Lifetimes in structs		20	40%
2		Rust Principles: Scope and Ownership, Visibility rules, Stack vs Heap, Binary Heap, Functions, Function parameters, Function as Parameters, Closures, Closures as Data structures, Functional Iteration, Collect, Enumerate, Map, Filter, Move, Clone, Copy, references-dereference and Borrowing		15	30%
3		Traits and Generics: Lifetimes, Annotating lifetimes, Type Aliasing, Import Aliasing Pattern Matching, Enums, Monads and Option, Control flow (If/else, If let, match, loops), Iterators, Collections, Sets Error Handling, Rust Testing, Rust File Handling, Rust Thread		10	30%
4		Practical Programs based on Control structures Programs based on Functions Programs based on Error Handling Programs based on File Handling - Rust Thread Programs		30	25%
Text Book					
1		The Rust Programming Language, 2nd Edition by Steve Klabnik, 2 nd Edition, 2023			
2		Programming with Rust by Donis Marshall, 1 st Edition, 2023			
Reference Books					
1	The Complete Rust Programming Reference guide by Rahul Sharma, 2019				

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Minor Discipline Specific Elective- II BCAXXXX: Introduction to Statistical Analysis Using R		Credits / Hours per week	4 (L-3, P-1)
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:		Maximum Marks / Grade	100
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit		Contact Hours	Weightage(%)
1		Introduction to R- A programming language and environment for data analysis and graphics. Syntax of R expressions: names and assignments, Operators and their precedence, subscripting, Data classes, modes and types. Data objects: Basic data objects- raw data, data in tabular form, matrices, lists, factors and ordered factors; Built-in-Functions- Elementary functions and summary functions, applying functions to subsets of data. Data frames: The benefits of data frames, creating data frames, combining data frames. Reading data with scan function. Arithmetic, logical and relational operators. Use of functions from “apply” family (apply, tapply, sapply etc.)		20	40%
2		Performing data analysis tasks: computing descriptive statistics, exploring data using graphical tools, interactively adding information of plot, multiple plot layout, interactively adding information to plot using functions – identify the plotted points, adding new data to current plot, adding text and legend etc.; graphics using “ggplot2” package. Exploring relationships between data: plotting scatter diagrams, correlation, simple and multiple linear regression analysis using R, adding fitted line to the scatter plot. Mathematical computing in R: probability and random numbers.		15	30%
3		Cross tabulating categorical data (pivot tables), cross classifying subsets of data frames, manipulating and analyzing cross classified data. Writing simple programs in R. Importing data files: import.data, read.table functions; Exporting data: cat, write, write.table functions. Use of source and sink functions.		10	30%
4		Practicals: Operators, functions and data classes Exploring data using graphical tools Mathematical computing		30	25%

	- Importing data files		
Reference Book			
1	Peter Dalgaard (2002): Introductory Statistics with R, Springer		
2	Tilman M. Davies (2016): The Book of R: A first course in programming and Statistics, William Pollock		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Skill Enhancement Course BCAXXXX: Innovation and Entrepreneurship	Credits / Hours per week	2 (L-2)	
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:	Maximum Marks / Grade	50	
Mode of Transaction		Lectures and Tutorials			
Unit No.		Topic/Unit	Contact Hours	Weightage(%)	
1		Introduction to Innovation and Entrepreneurship Overview of innovation and entrepreneurship, Historical context and significance, Key concepts and definitions, Entrepreneurship ecosystems Identifying Opportunities Recognizing market gaps and unmet needs, Conducting market research and analysis, Customer discovery and personas, Problem-solving and ideation Lean Startup Methodology Lean startup principles and practices, Minimum Viable Product (MVP) development, Validating business ideas through experimentation, Pivot and persevere strategies Business Model Canvas Business model components and value proposition, Revenue models and pricing strategies, Customer channels and acquisition, Cost structure and resource allocation Feasibility and Viability Analysis Financial forecasting and budgeting, Risk assessment and mitigation strategies, Legal and regulatory considerations Intellectual property and patents	15	50	
2		Launching and Scaling Funding options for startups (e.g., bootstrapping, venture capital), Scaling strategies and growth hacking, Building a strong company culture, Managing resources and operations	15	50	

	Leadership and Decision-Making Leadership skills and styles in entrepreneurship, Effective decision-making in startups, Ethical considerations in entrepreneurship, Case studies of successful entrepreneurs Pitching and Presentation Crafting a compelling pitch and elevator pitch, Pitching to investors and stakeholders, Presentation skills and public speaking, Feedback and refinement Entrepreneurship Ecosystems Incubators, accelerators, and startup support networks, Collaborative and co-working spaces, Government initiatives and policies, Networking and mentorship opportunities Emerging Trends and Future of Entrepreneurship Innovations in entrepreneurship (e.g., social entrepreneurship, sustainable entrepreneurship), Technology trends impacting entrepreneurship		
Text Book			
1	Innovation and Entrepreneurship: Practice and Principles, Peter F. Drucker, Harper Business, Revised Edition, 2007		
Reference Books			
1	Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses, Eric Ries,Currency, 2011		
2	The Innovator's Dilemma: When New Technologies Cause Great Firms to Fail, Clayton Christensen, Harvard Business Review Press, 2013		

	The Maharaja Sayajirao University of Baroda Faculty of Science Department of Computer Applications			Academic Year	2025-26
Bachelor of Computer Applications (Honors)					
Year	III	Skill Enhancement Course BCAXXXX: Digital Marketing	Credits / Hours per week	2 (L-1, P-1)	
Semester	V	Year of Introduction: 2025 onwards Year of Syllabus Revision:	Maximum Marks / Grade	50	
Mode of Transaction		Lectures, Tutorials and Practical			
Unit No.		Topic/Unit	Contact Hours	Weightage(%)	
1		INTRODUCTION TO DIGITAL MARKETING The Digital Marketing Institute Method: Definition of the DMI Method, Principles, Tools, The DMI Framework, The DMI Quality Scale Digital Marketing: Key Concepts of Digital Marketing, Traditional v. Digital Marketing, The Opportunity of Digital Marketing, Characteristics of Digital Marketing, Implications of Digital Marketing, Market Research. Market Reality SEARCH MARKETING (SEO) SEARCH ENGINE OPTIMIZATION Search Results & Positioning, Benefits of Search Position, Stakeholders in Search, Mechanics of Search, On Page Optimisation, The SEO Process, Customer Insights, Analysis & Review, Keyword Research & Selection, Content Updates & Layout, Meta Tags, SEO Site Map, SEO Google Search Console, Off-Page Optimisation, Inbound Links & Link Building, Ranking SEARCH MARKETING (PPC) PAY PER CLICK Key PPC Concepts: Strengths of Pay Per Click, Keyword Research, Google PPC, Research Tools, Search Campaign Process, Keyword Selection, Ad Copy, Landing Pages, Targeting, Budgets, Scheduling, Display Networks, Ad Centre, Campaign Management, Conversion Tracking, DIGITAL DISPLAY ADVERTISING Benefits of Digital Display, Challenges of Digital Display, Business Value, Running Effective Ads, Ad Formats- Features, Ad Display Frequency, Campaign Objectives, Budget, Planning and Steps, Target Audience, Creative Formats, Targeting, Tracking your Campaign EMAIL MARKETING Online Data Capture, Offline Data Capture, Segmentation, Email Design, User Behaviour, User Characteristics, Email Copy, Email Structure, Email Delivery, Email Systems, Filtering, Scheduling, Measurement, Key Terms & Metrics, Split Testing	15	50	

2	Practicals: • Practical based on the above topic • Study or case study of different websites regarding their Digital Marketing strategy	30	50
Text Book			
1	Digital Marketing: Strategy, Implementation & Practice by Dave Chaffey & Fiona Ellis-Chadwick, 8 th edition, 2022, Pearson		
2	Understanding Digital Marketing: Marketing Strategies for Engaging the Digital Generation by Damian Ryan, 4 th edition, 2017		